

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, JavaScript, TypeScript, Java, C++, SQL, Bash

Backend & APIs

FastAPI, Node.js, Express, REST APIs, async services, multi-tenant systems, authentication, rate limiting, integration testing

Databases & Data

PostgreSQL, MongoDB, Redis, SQLite, Apache Spark / PySpark, data pipelines, aggregation queries

Cloud & DevOps

AWS (S3, KMS), GCP, Azure, Docker, Git, GitHub Actions, CI/CD, git-webhook auto-deploy, Linux

AI & Tooling

Claude / Claude Code, GitHub Copilot, Cursor, OpenAI / Groq APIs, LLM agents, RAG, PyTorch

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Architected full-stack data annotation platform (**React, Node.js, MongoDB**) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved labeling throughput by **60%**.
- Owned backend reliability across the eTriage stack: monitored production logs during feature releases and client integrations, shipped fixes for tenant-specific incidents, hardened authentication and rate limiting, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive clinical data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Developed multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, classifies evidence regimes, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and reviewer-facing trust user interface (UI) for inspecting extraction quality and ranking transparency.

Spark ETL - Backend Data Processing at Scale

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Built a backend data pipeline over a **1.4B-row** dataset, cutting end-to-end runtime **25%** via broadcast joins, adaptive query execution, and partition tuning to scale efficiently.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition/column pruning and filter pushdown, profiling stage timings via Spark UI.

cf-ai-research-scout - Edge-Native AI Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, WebSockets, SQL

- Built edge-native AI research assistant: agent fetches web sources, generates multi-step research digests, and streams responses back to the client over **WebSocket** in real time, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.